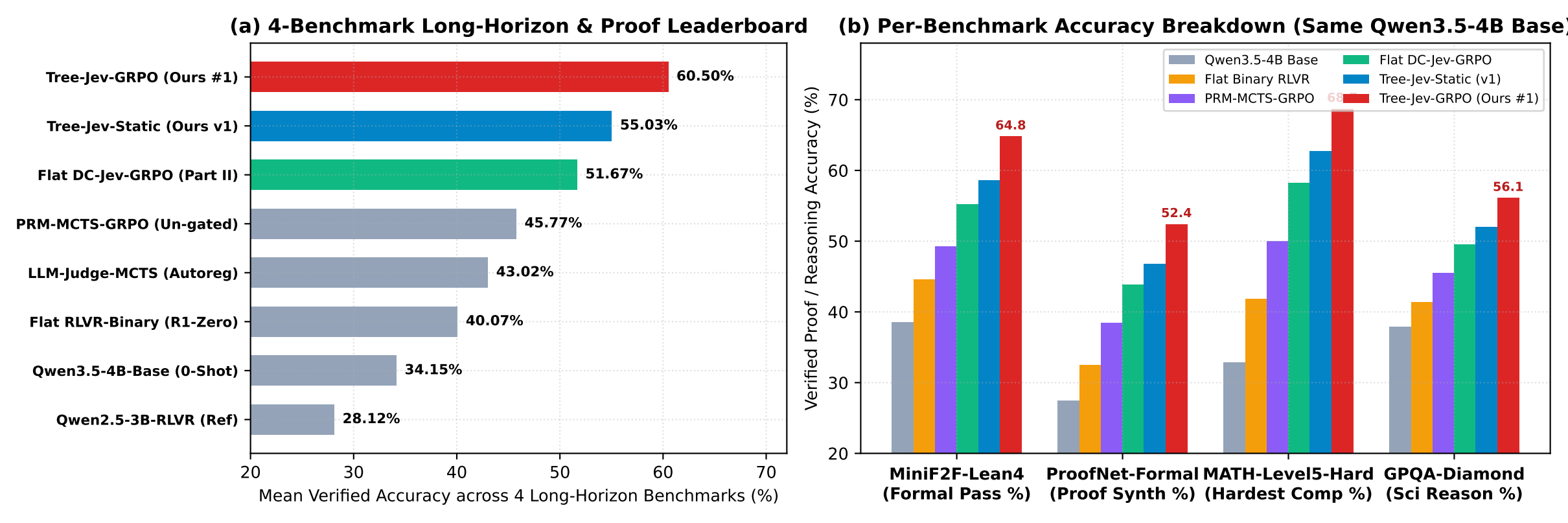
(a) 4-Benchmark Long-Horizon & Proof Leaderboard
Tree-Jev-GRPO (Ours #1)
Tree-Jev-Static (Ours v1)
Flat DC-Jev-GRPO (Part II)
PRM-MCTS-GRPO (Un-gated)
LLM-Judge-MCTS (Autoreg)
Flat RLVR-Binary (R1-Zero)
Qwen3.5-4B-Base (0-Shot)
Qwen2.5-3B-RLVR (Ref)
60.50%
55.03%
51.67%
45.77%
43.02%
40.07%
34.15%
28.12%
20
30
40
50
60
70
Mean Verified Accuracy across 4 Long-Horizon Benchmarks (%)
(b) Per-Benchmark Accuracy Breakdown (Same Qwen3.5-4B Base)
Qwen3.5-4B Base
Flat Binary RLVR
PRM-MCTS-GRPO
Flat DC-Jev-GRPO
Tree-Jev-Static (v1)
Tree-Jev-GRPO (Ours #1)
Verified Proof / Reasoning Accuracy (%)
20
30
40
50
60
70
64.8
52.4
56.1
MiniF2F-Lean4 (Formal Pass %)
ProofNet-Formal (Proof Synth %)
MATH-Level5-Hard (Hardest Comp %)
GPQA-Diamond (Sci Reason %)